

PolicyMind

How to run your Dealer Selection Guidelines chatbot — a step-by-step guide for restarting the application on your own machine.

What you are running

A fully local, free-to-run RAG chatbot over the PSU OMC *Dealer Selection Guidelines (June 2023)*. It uses BGE embeddings + ChromaDB + BM25 for retrieval, a cross-encoder for reranking, and Groq's free Llama 3.3 endpoint for answers. Everything except the LLM call runs on your machine; no documents leave your computer.

Prerequisites (one-time)

- **Python 3.10 or newer** already installed (you have Python 3.13).
- **Project folder** at `D:\PERSONAL_PROJECTS\POLICYMIND\RETRIEVAL AND CHUNKING WITH CLAUDE CODE\chatbot\`.
- **A free Groq API key** — see Step 1 below.
- **~3 GB free disk space** for the embedding + reranker models (already downloaded if you have run it once).

Steps to run the chatbot

1 Get a free Groq API key (one-time only)

Groq provides a generous free tier for Llama 3.3 70B — no credit card required.

- 1 Open <https://console.groq.com> in your browser.
- 2 Sign in with Google or email.
- 3 Go to **API Keys** → click **Create API Key**.
- 4 Copy the key (starts with gsk_...) and save it somewhere safe.

Note You only need to do this once. The same key works every time you launch the chatbot.

2 Open PowerShell in the project folder

Press **Win + R**, type **powershell**, press Enter. Then run:

```
cd "D:\PERSONAL_PROJECTS\POLICYMIND\RETRIEVAL AND CHUNKING WITH CLAUDE CODE\chatbot"
```

3 Launch the chatbot

Run the Streamlit app via Python:

```
python -m streamlit run app.py
```

Streamlit will print a URL like `http://localhost:8501` and your browser should open automatically. If it doesn't, copy that URL into Chrome / Edge yourself.

Note First launch after a reboot may take 30-60 seconds while the embedding and reranker models load into memory. Subsequent queries are fast.

4 Paste your Groq API key in the sidebar

In the left sidebar, paste your gsk_... key into the **Groq API Key** box. The key is held in memory only — it is not saved to disk.

Note Want to skip pasting every time? Create a file named `.env` in the chatbot folder with one line: `GROQ_API_KEY=gsk_your_key_here`. Streamlit will pick it up automatically on next launch.

5 Ask a question

Either click one of the **Quick Questions** buttons in the sidebar, or type your own question in the chat box at the bottom of the page. Examples:

- *What are the eligibility criteria for an SC/ST individual applicant?*
- *What is the non-refundable application fee for a rural OBC location?*
- *Explain the online draw of lots procedure.*
- *What disqualifies an applicant?*
- *What infrastructure must a CC Rural retail outlet have?*

Every answer cites the **Section number** and **page number** from the source PDF. Click **View Sources** below the answer to see exactly which chunks were used.

6 Stop the chatbot when you are done

Go back to the PowerShell window where Streamlit is running and press **Ctrl + C**. The browser tab can simply be closed.

Note Closing the browser tab does NOT stop the server. You must press Ctrl+C in PowerShell.

Troubleshooting

'streamlit' is not recognized	Always launch via <code>python -m streamlit run app.py</code> . The bare <code>streamlit</code> command isn't on your PATH.
Browser shows 'Index not found'	Re-build the search index once: <code>python ingest.py</code> . Takes 1-2 minutes.
Groq API error / rate limit	The free tier allows 30 requests/min and 14,400 requests/day. Wait a minute and retry. If your key is invalid, regenerate it at console.groq.com .
Port 8501 already in use	Another Streamlit is still running. Either close it via Ctrl+C in its window, or launch on a different port: <code>python -m streamlit run app.py --server.port 8502</code> .
Models re-downloading every launch	The HuggingFace cache lives at <code>C:\Users\<you>\.cache\huggingface</code> . Don't delete it — that's where the ~750 MB of models are stored.

What lives where

File	What it does
app.py	The Streamlit user interface.
retrieval.py	Hybrid dense + sparse search, reranking, parent-promotion.
ingest.py	PDF → chunks → ChromaDB + BM25. Only re-run if the PDF changes.
config.py	Model names, paths, section titles, abbreviation glossary.
chroma_db/	Persistent vector index (~5 MB).
bm25_index.pkl	Persistent BM25 keyword index.

Generated for internal use. Policy source: Dealer Selection Guidelines, June 2023, Version 1.0/10.06.2023.